Cumulative Distribution Function

PMF $P_X(x_k)=\frac{1}{4}$ PMF discrete random variable

↳ CDF can be used for any kind of random variable

-Definition

A CDF of a random variable X is defined as $F_X(x)=P(X\le x)$ for all $x\in\mathbb{R}$

EX: I have tossed a coin twice. RV X is number of heads. Find its PMF and CDF.

$R_X=\{0,1,2\}$ $\underbrace{P_X(0)=\frac{1}{4} \quad P_X(1)=\frac{1}{2} \quad P_X(2)=\frac{1}{4}}_{PMF}$

CDF: $F_X(x)=P(X\le x)$

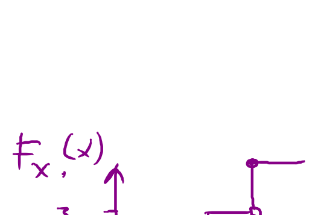
For $\alpha<0$ $F_X(\alpha)=0$

$\alpha\ge 2$ $F_X(\alpha)=1=P_X(0)+P_X(1)+P_X(2)$

$0\le\alpha<1$ $F_X(\alpha)=\frac{1}{4}$

$1\le\alpha<2$ $F_X(\alpha)=\frac{3}{4}=P_X(0)+P_X(1)$

$$F_X(x)=\begin{cases}0 & \alpha<0\\ \frac{1}{4} & 0\le\alpha<1\\ \frac{3}{4} & 1\le\alpha<2\\ 1 & 2\le\alpha\end{cases}$$



→ CDF

Properties of Cumulative Distribution Function

1-) $F_X(-\infty)=0$ $F_X(\infty)=1$

2-) CDF is a $\underbrace{\text{non decreasing}}_{\text{azalmayan}}$ function

$\alpha\le\beta \Rightarrow F_X(\alpha)\le F_X(\beta)$

3-) For any $x_k\in R_X$

$F_X(x_k)-F_X(x_k-\underset{\text{epsilon}}{\epsilon})=P_X(X=x_k)$

4-) $P(a<x\le b)=P_X(X\le b)-P_X(X\le a)=F_X(b)-F_X(a)$

EXPECTATION

$a_1,a_2 \dots a_n$ Consider a R.V X, how we find its average?

↓ average $\underbrace{\sum_{i=0}^{n} a_i}_{\text{mean average}}$

Let X be a R.V. with $R_X=\{x_1,x_2,\dots\infty\}$ the expected value of random variable X, EX

$$EX=\sum_{x_k\in R_X} x_k\,.\,P(X=x_k)$$

Expected value

$E[X]=E(X)=\mu X$

EX:

x	P(x)
2	0.75
3	0.25

EX of X $=\sum x_k . P_X(x_k)$

$=2.0{,}75+3.0{,}25$

$=2.25\to EX$

FUNCTIONS OF RANDOM VARIABLES

X is a random variable $R_Y=\{g(x)\mid x\in R_X\}$

$\underset{\text{R.V.}}{Y}=g(X)\xrightarrow{\text{function}}$ R.V.

we already know the PMF(X), PMF(Y)

$P_Y(y)=P(Y=y)=P(g(x)=y)=\sum_{x:g(x)=y}P_X(x)$

EX: Let X be a discrete R.V. with $P_X(k)=\frac{1}{5}$ for $k=-1,0,1,2,3$

Let $Y=2|X|$ find range and pmf of Y.

$R_X=\{-1,0,1,2,3\}$ $P_X(0)=P_X(1)\dots P_X(3)$ $R_Y=\{|2.-1|,|2.0|,|2.1|,|2.2|,|2.3|\}$ ↳ $\{2,0,4,6\}$

$P_Y(0)=\frac{1}{5}$ $P_Y(2)=\frac{2}{5}$ $P_Y(4)=\frac{1}{5}$ $P_Y(6)=\frac{1}{5}$

↳ $P_X(-1)+P_X(1)=\frac{2}{5}$

Law of Unconscious statistician (LOTUS) for D.R.V.

$$\underbrace{E[g(X)]}_{Y}=\sum_{x_k\in R_X} g(x_k).\,P_X(x_k)$$

EX. prev. question → Find E[Y] where $Y=2|X|$

LOTUS $E[Y]=2|-1|.\frac{1}{5}+2.|0|.\frac{1}{5}+2.|1|.\frac{1}{5}+2.|2|.\frac{1}{5}+2.|3|.\frac{1}{5}=\frac{14}{5}=\mu_{E(Y)}$

σ^2 → Variance and σ → standard deviation

- I am offered to invest $800, 2 investment accounts.

1- will give me $1000 in a year.

2- will give me either 500$ or $1500 (equally likely) in one year.

which one should I choose?

$P_X(1000)=1$ ↳ $EX=1000$ $P_Y(500)=\frac{1}{2}$ $P_Y(1500)=\frac{1}{2}$

$EX=500.\frac{1}{2}+1500.\frac{1}{2}=1000$ → Same

Only expectation is not sufficient → sadece E yeterli değil

The variance of a RV X with $EX=\alpha_X$ $Var(X)=E[(X-\underbrace{\mu_X}_{\text{mean}})^2]$

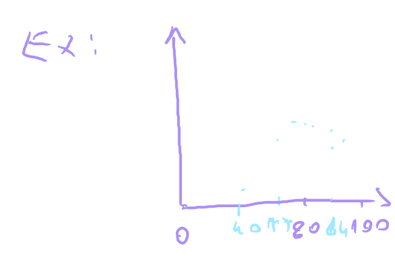
↳ $Var(X)=1000-1000=0$

↳ $Var(Y)=\underbrace{500-1000}_{(-500)^2}+1500-1000$ "σ^2" ?

Standard deviation: is a square root of variance

$\sigma=std(X)=\sqrt{Var(X)}$

EX:



100 students

mean → 78

variance-small

Covariance and Correlation

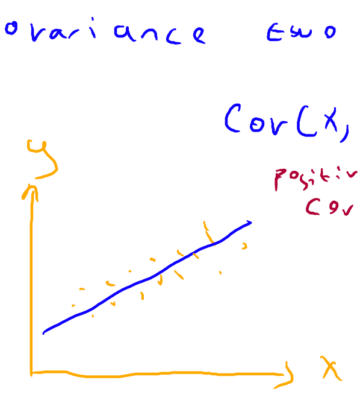
Ex, var, std. dev → distribution of a single R.V.

→ Two R.V. → covariance, correlation

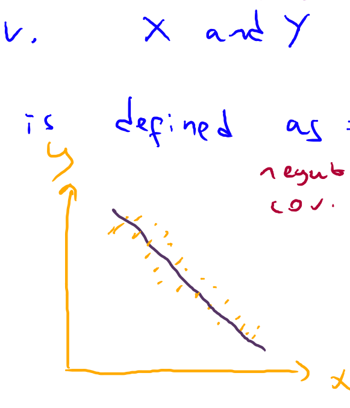
→ Covariance measure the association of two R.V.

Covariance two R.V. X and Y

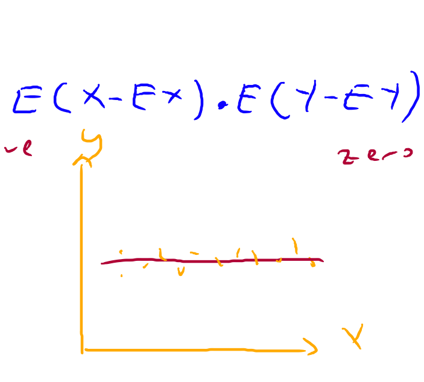
$Cov(X,Y)$ is defined as $=E(X-EX).E(Y-EY)=E(XY)-E(X).E(Y)$



$Cov(X,Y)>0$ both increasing



$Cov(X,Y)<0$ ↳ One of x,y is inc. and the other one is decreasing



$Cov(X,Y)=0$ → no relation between x,y

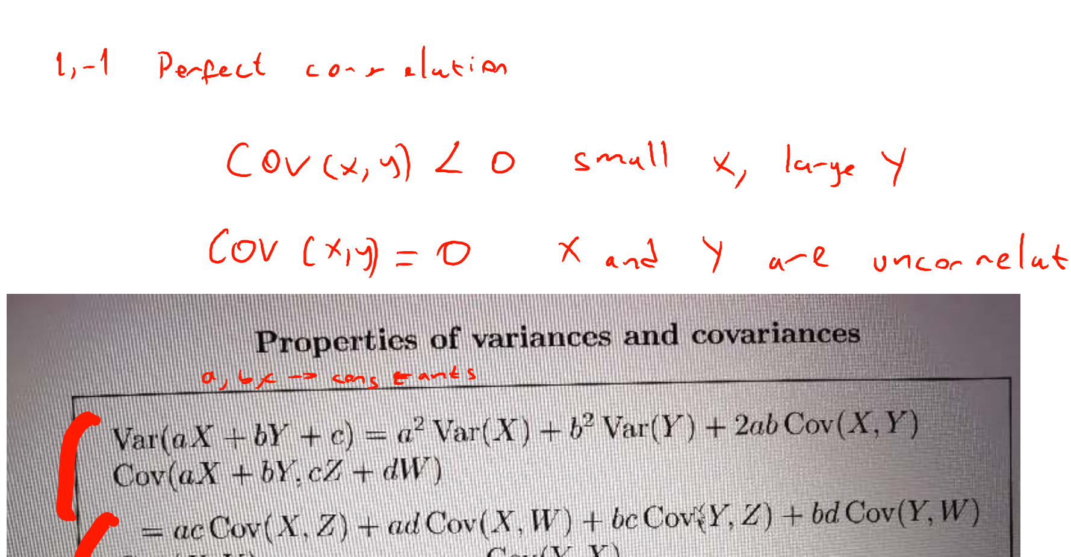
Correlation ↓ like covariance, the values of correlation is between [-1;1]

"ρ" ro $=\frac{Cov(X,Y)}{std(X).std(Y)}=\frac{cov(x,y)}{\sigma_x.\sigma_y}$

1,-1 Perfect correlation

$Cov(x,y)<0$ small x, large Y

$Cov(X,Y)=0$ X and Y are uncorrelated



Properties of variances and covariances

a, b, c → constants

$Var(aX+bY+c)=a^2Var(X)+b^2Var(Y)+2abCov(X,Y)$

$Cov(aX+bY,cZ+dW)$

$=acCov(X,Z)+adCov(X,W)+bcCov(Y,Z)+bdCov(Y,W)$

$Cov(X,Y)=Cov(Y,X)$

$\rho(X,Y)=\rho(Y,X)$

In particular,

$Var(aX+b)=a^2Var(X)$

$Cov(aX+b,cY+d)=acCov(X,Y)$

$\rho(aX+b,cY+d)=\rho(X,Y)$

For independent X and Y,

$Cov(X,Y)=0$

$Var(X+Y)=Var(X)+Var(Y)$

Ex: A program consists two modules. The number of errors in the first module X and the number of errors in the second module Y have joint distribution.

$P(x,y)$ $P(0,0)=P(0,1)=P(1,0)=0.2$

$P(1,1)=P(1,2)=P(1,3)=0.1$

$P(0,2)=P(0,3)=0.05$

$$x\begin{cases}\begin{array}{c|cccc} & 0 & 1 & 2 & 3\\ \hline 0 & 0.2 & 0.2 & 0.05 & 0.05\\ 1 & 0.2 & 0.1 & 0.1 & 0.1\end{array}\end{cases}\quad y$$

a) marginal Prob. X and Y P(X), P(Y)

$P_x(0)=\sum_y P(0,y)=0.2+0.2+0.05+0.05=0.5$ $P_x(1)=\sum_y P(1,y)=0.2+0.1+0.1+0.1=0.5$

$P_y(0)=\sum_x P(x,0)=0.2+0.2=0.4$ $P_y(1)=\sum_x P(x,1)=0.2+0.1=0.3$ $P_y(2)=\sum_x P(x,2)=0.1+0.05=0.15$

$P_y(3)=\sum_x P(x,3)=0.15$

b) X and Y are independent?

$P(x,y)=P(x).P(y)$

$P_{X,Y}(0,1)=P_x(0).P_y(1)$

$0.2=?\ 0.5.0.5$ No → X and Y aren't independent

c) Var(X)? Var(Y)=?

$Var(X)=E[(X-\mu_x)^2]$ $E(X)=\alpha_x=\sum_{x=0}^{1}x.P_x=0.0.5+1.0.5=0.5$

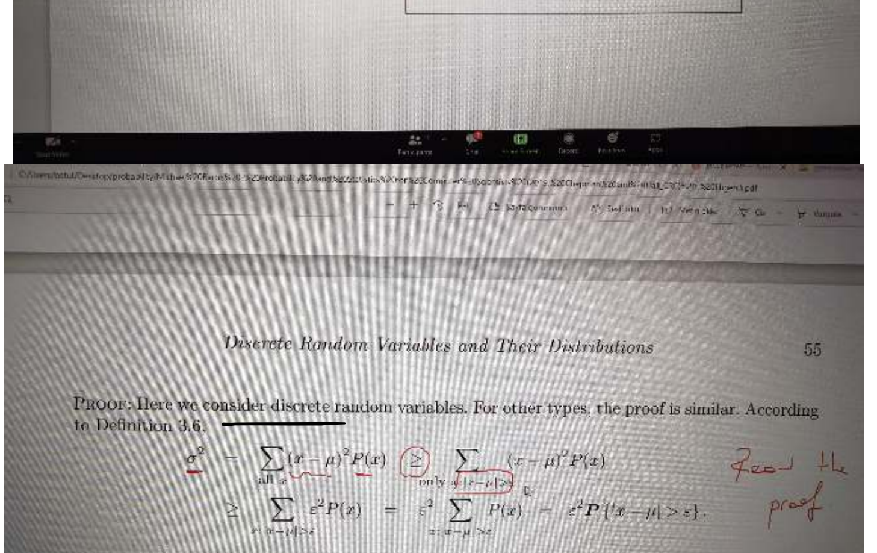
$Var(Y)=E[(Y-\mu_Y)^2]$ $E(Y)=\alpha_y=\sum_{y=0}^{3}y.P_y=0\times0.4+1.0.3+2\times0.15+3\times0.15=1{,}05$

$Var(X)=E[(X-\mu_X)^2]=\sum_{x=0}(0-0.5)^2+(1-0.5)^2=5$

$Var(Y)=\sum_{y=0}^{y=3}\to(0-1.05)^2+(1-1.05)^2+(2-1.05)^2+(3-1.05)^2$

d) $P(X,Y)=?$ → covariance $\rho_{X,Y}=cov(X,Y)=E(X-\overset{0.5}{\alpha_x})(Y-\overset{1.05}{\mu_y})$

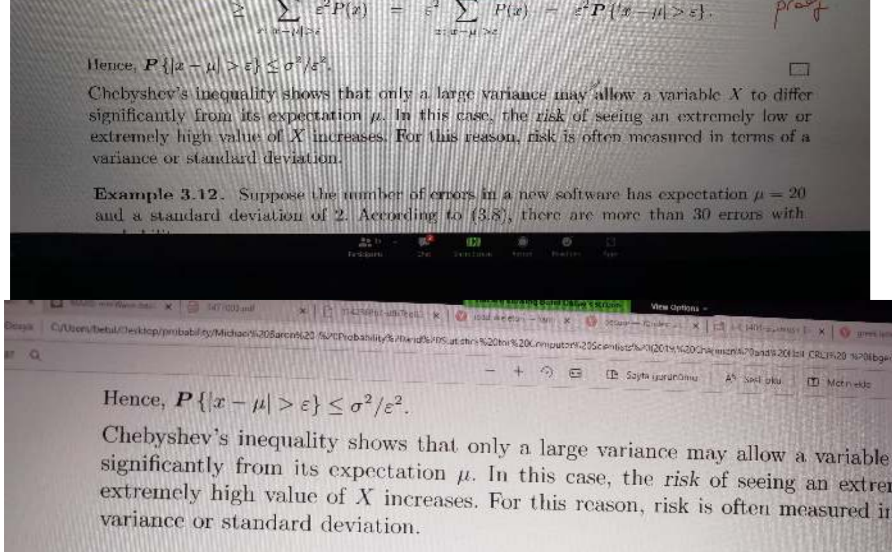
$=E(X.Y)-E(X).E(Y)$



3.3.7 Chebyshev's inequality

Knowing just the expectation and variance, one can find the range of values most likely taken by this variable. Russian mathematician *Pafnuty Chebyshev* (1821–1894) showed that any random variable X with expectation $\overset{\text{mean}}{\mu}=E(X)$ and variance $\sigma^2=Var(X)$ belongs to the interval $\mu\pm\varepsilon=[\mu-\varepsilon,\mu+\varepsilon]$ with probability of at least $1-(\sigma/\varepsilon)^2$. That is,

Chebyshev's inequality $P\{|X-\mu|>\varepsilon\}\le\left(\frac{\sigma}{\varepsilon}\right)^2$ for any distribution with expectation μ and variance σ^2 and for any positive ε. (3.8)



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PROOF: Here we consider discrete random variables. For other types, the proof is similar. According to Definition 3.6,

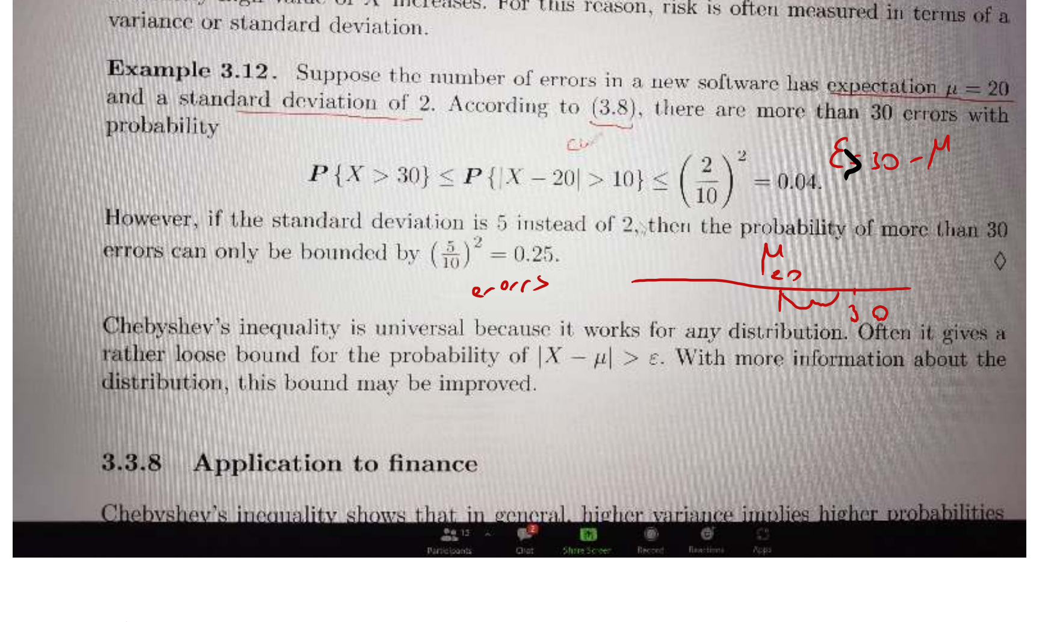
$$\sigma^2=\sum_{\text{all }x}(x-\mu)^2P(x)\ \ge\sum_{\text{only }x:|x-\mu|>\varepsilon}(x-\mu)^2P(x)\ \ge\sum_{x:|x-\mu|>\varepsilon}\varepsilon^2P(x)=\varepsilon^2\sum_{x:|x-\mu|>\varepsilon}P(x)=\varepsilon^2P\{|x-\mu|>\varepsilon\}.$$

Read the proof

Hence, $P\{|x-\mu|>\varepsilon\}\le\sigma^2/\varepsilon^2$. ☐

Chebyshev's inequality shows that only a large variance may allow a variable X to differ significantly from its expectation μ. In this case, the *risk* of seeing an extremely low or extremely high value of X increases. For this reason, risk is often measured in terms of a variance or standard deviation.

Example 3.12. Suppose the number of errors in a new software has expectation μ = 20 and a standard deviation of 2. According to (3.8), there are more than 30 errors with



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Example 3.12. Suppose the number of errors in a new software has expectation μ = 20 and a standard deviation of 2. According to (3.8), there are more than 30 errors with probability

$$P\{X>30\}\le P\{|X-20|>10\}\le\left(\frac{2}{10}\right)^2=0.04.$$

$\varepsilon>30-\mu$

However, if the standard deviation is 5 instead of 2, then the probability of more than 30 errors can only be bounded by $\left(\frac{5}{10}\right)^2=0.25$. ◊

errors μ 20 30

Chebyshev's inequality is universal because it works for any distribution. Often it gives a rather loose bound for the probability of $|X-\mu|>\varepsilon$. With more information about the distribution, this bound may be improved.

3.3.8 Application to finance

Chebyshev's inequality shows that in general, higher variance implies higher probabilities

look cheby shows in equality from bras book

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